

ELECTRON-PHONON INTERACTION IN METALS

Deformation potential

$$\hat{H}_{\text{e-ph}} = -\Xi \vec{\nabla} \cdot \hat{\mathbf{u}}$$

$$\hat{\mathbf{u}} = \sum_{\mathbf{q}} \sqrt{\frac{\hbar}{2\rho V \omega_{\mathbf{q}}}} \left(e^{i(\mathbf{q} \cdot \mathbf{r} - \omega_{\mathbf{q}} t)} \hat{c}_{\mathbf{q}} + e^{-i(\mathbf{q} \cdot \mathbf{r} - \omega_{\mathbf{q}} t)} \hat{c}_{\mathbf{q}}^{\dagger} \right)$$

$$\hat{H}_{\text{e-ph}} = \sum_{\mathbf{q}} \sqrt{\frac{\hbar}{2\omega_{\mathbf{q}} \rho V}} i\Xi q \left(\underbrace{e^{i(\mathbf{q} \cdot \mathbf{r} - \omega_{\mathbf{q}} t)} \hat{c}_{\mathbf{q}}}_{\text{absorption}} - \underbrace{e^{-i(\mathbf{q} \cdot \mathbf{r} - \omega_{\mathbf{q}} t)} \hat{c}_{\mathbf{q}}^{\dagger}}_{\text{emission}} \right)$$

Matrix elements

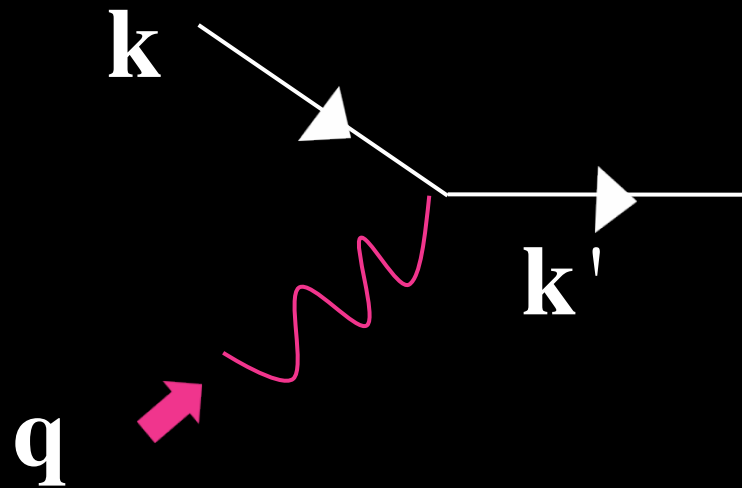
$$\psi_{\mathbf{k}} = \frac{e^{i(\mathbf{k} \cdot \mathbf{r} - \varepsilon_{\mathbf{k}} t / \hbar)}}{\sqrt{V}} \Rightarrow \psi_{\mathbf{k}'} = \overset{\text{absorption}}{\frac{e^{i(\mathbf{k}' \cdot \mathbf{r} - \varepsilon_{\mathbf{k}'} t / \hbar)}}{\sqrt{V}}}$$

$$n_{\mathbf{q}} \Rightarrow n_{\mathbf{q}} - 1$$

$$M_{\mathbf{k}\mathbf{k}'} = \frac{1}{V} \sqrt{\frac{\hbar}{2\rho\omega_{\mathbf{q}}V}} iq\Xi \underbrace{\int d^3r e^{-i\mathbf{k}' \cdot \mathbf{r}} e^{i\mathbf{q} \cdot \mathbf{r}} e^{i\mathbf{k} \cdot \mathbf{r}}}_{V\delta_{\mathbf{k}', \mathbf{k}+\mathbf{q}}} \delta(\varepsilon_{\mathbf{k}'} - \varepsilon_{\mathbf{k}} - \hbar\omega_{\mathbf{q}}) \\ \times \underbrace{\langle \langle n_{\mathbf{q}} - 1 | c | n_{\mathbf{q}} \rangle \rangle}_{\sqrt{N_{\mathbf{q}}}}$$

$$N_{\mathbf{q}} = \frac{1}{e^{\hbar\omega_{\mathbf{q}}/k_B T} - 1}$$

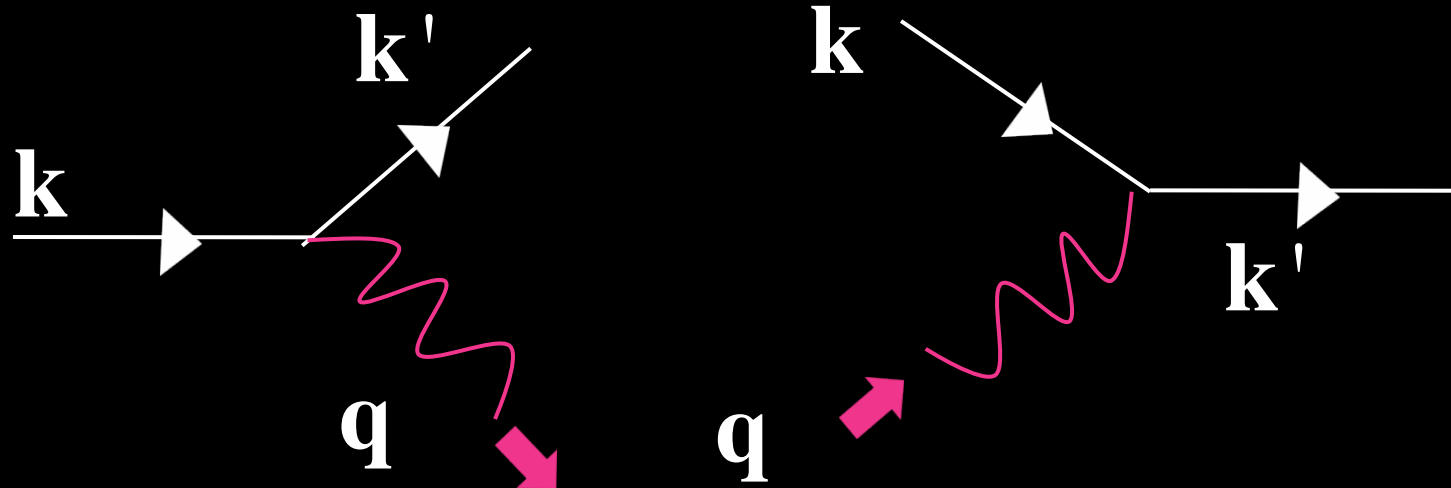
Scattering rate



$$W_{\mathbf{k} \rightarrow \mathbf{k}'}^{\text{abs}} = \frac{2\pi}{\hbar} \frac{\hbar \Xi^2 q^2}{2\rho\omega_{\mathbf{q}}V} N_{\mathbf{q}} f_{\mathbf{k}} (1 - f_{\mathbf{k}'})$$

Scattering processes

Loss:



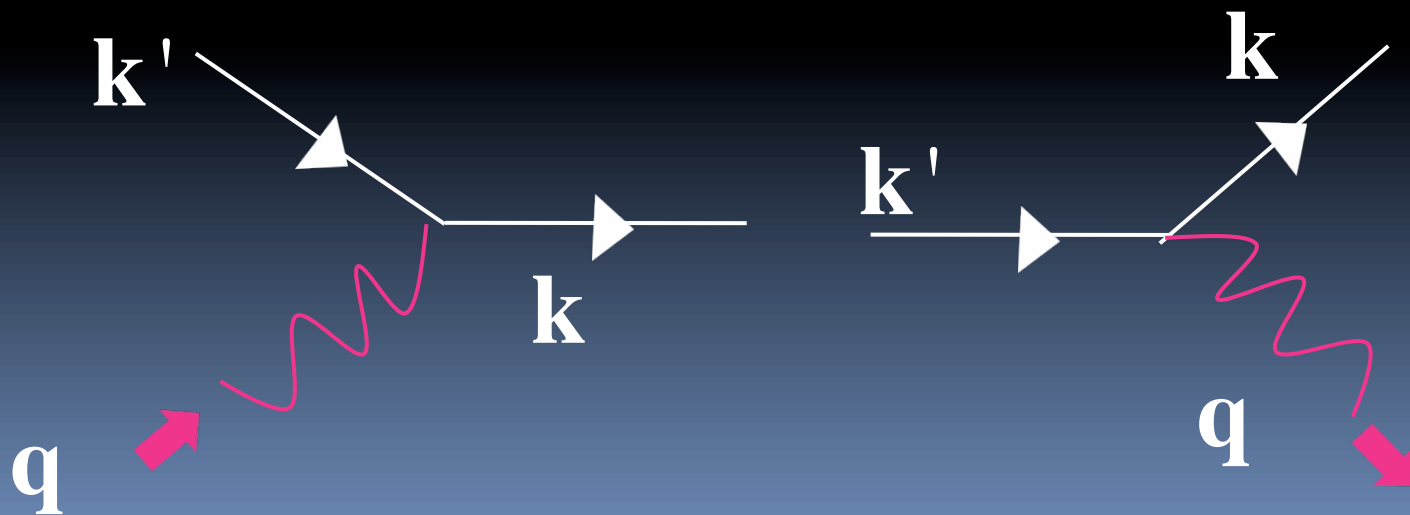
$$k' = k - q$$

$$\varepsilon_{k-q} + \hbar\omega_q = \varepsilon_k$$

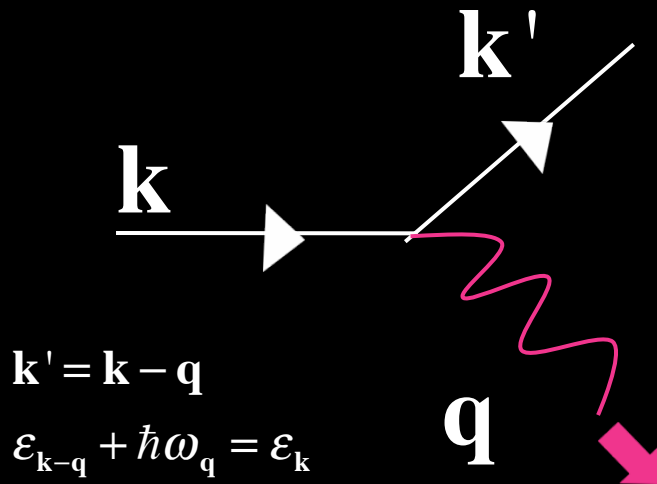
$$k' = k + q$$

$$\varepsilon_k + \hbar\omega_q = \varepsilon_{k+q}$$

Gain:

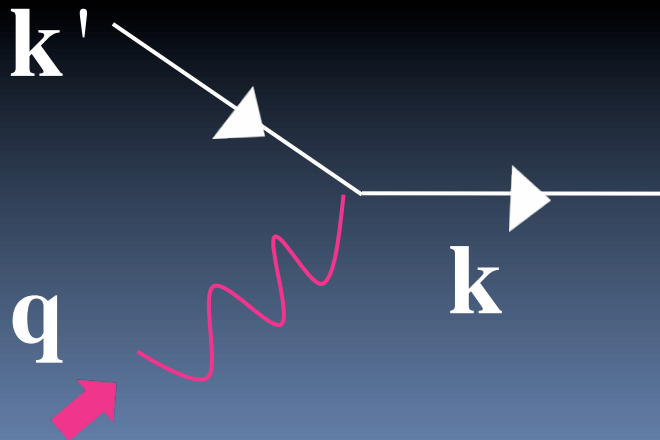


Constructon of the collision integral



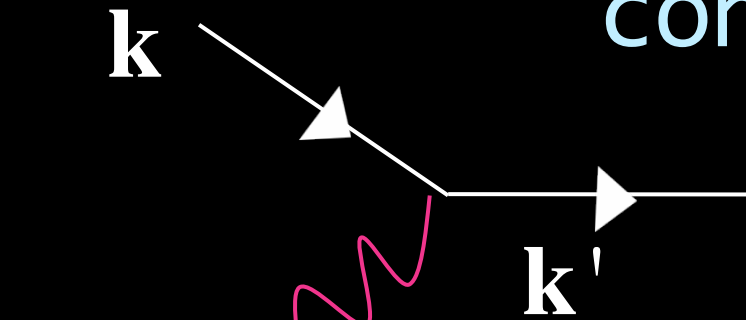
$$I_1 = 2\pi \int \frac{d^3q}{(2\pi)^3} \frac{\Xi^2 q^2}{2\rho\omega_{\mathbf{q}}} \delta(\varepsilon_{\mathbf{k}-\mathbf{q}} + \hbar\omega_{\mathbf{q}} - \varepsilon_{\mathbf{k}}) \\ \times [N_{\mathbf{q}} f_{\mathbf{k}-\mathbf{q}} (1 - f_{\mathbf{k}}) - (N_{\mathbf{q}} + 1) f_{\mathbf{k}} (1 - f_{\mathbf{k}-\mathbf{q}})]$$

$\mathbf{q} \rightarrow -\mathbf{q}; \omega_{\mathbf{q}} = \omega_{-\mathbf{q}}; \text{equilibrium phonons: } N_{-\mathbf{q}} = N_{\mathbf{q}}$



$$I_1 = 2\pi \int \frac{d^3q}{(2\pi)^3} \frac{\Xi^2 q^2}{2\rho\omega_{\mathbf{q}}} \delta(\varepsilon_{\mathbf{k}+\mathbf{q}} + \hbar\omega_{\mathbf{q}} - \varepsilon_{\mathbf{k}}) \\ \times [N_{\mathbf{q}} f_{\mathbf{k}+\mathbf{q}} (1 - f_{\mathbf{k}}) - (N_{\mathbf{q}} + 1) f_{\mathbf{k}} (1 - f_{\mathbf{k}+\mathbf{q}})]$$

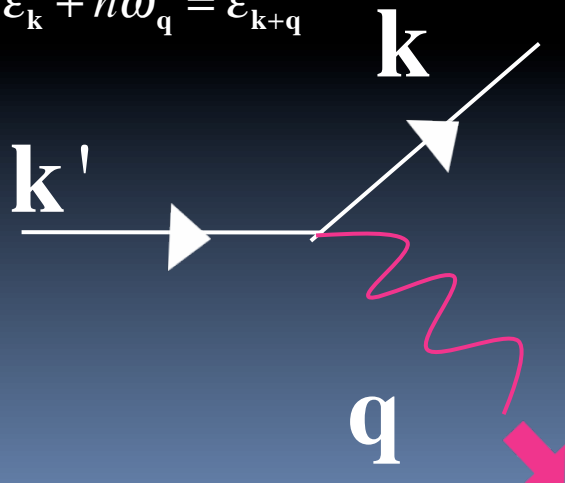
Collision integral, continued



$\mathbf{q}$

$$\mathbf{k}' = \mathbf{k} + \mathbf{q}$$

$$\varepsilon_{\mathbf{k}} + \hbar\omega_{\mathbf{q}} = \varepsilon_{\mathbf{k}+\mathbf{q}}$$



$$I_2 = 2\pi \int \frac{d^3q}{(2\pi)^3} \frac{\Xi^2 q^2}{2\rho\omega_{\mathbf{q}}} \delta(\varepsilon_{\mathbf{k}+\mathbf{q}} - \hbar\omega_{\mathbf{q}} - \varepsilon_{\mathbf{k}}) \\ \times \left[(N_{\mathbf{q}} + 1) f_{\mathbf{k}+\mathbf{q}} (1 - f_{\mathbf{k}}) - N_{\mathbf{q}} f_{\mathbf{k}} (1 - f_{\mathbf{k}+\mathbf{q}}) \right]$$

Combining the two parts together

$$\begin{aligned} & 2\pi \int \frac{d^3 q}{(2\pi)^3} \frac{\Xi^2 q^2}{2\rho\omega_q} \delta(\varepsilon_{\mathbf{k}+\mathbf{q}} + \hbar\omega_q - \varepsilon_{\mathbf{k}}) \\ & \times \left[N_q f_{\mathbf{k}+\mathbf{q}} (1 - f_{\mathbf{k}}) - (N_q + 1) f_{\mathbf{k}} (1 - f_{\mathbf{k}+\mathbf{q}}) \right] \\ & + \\ & 2\pi \int \frac{d^3 q}{(2\pi)^3} \frac{\Xi^2 q^2}{2\rho\omega_q} \delta(\varepsilon_{\mathbf{k}+\mathbf{q}} - \hbar\omega_q - \varepsilon_{\mathbf{k}}) \\ & \times \left[(N_q + 1) f_{\mathbf{k}+\mathbf{q}} (1 - f_{\mathbf{k}}) - N_q f_{\mathbf{k}} (1 - f_{\mathbf{k}+\mathbf{q}}) \right] \end{aligned}$$

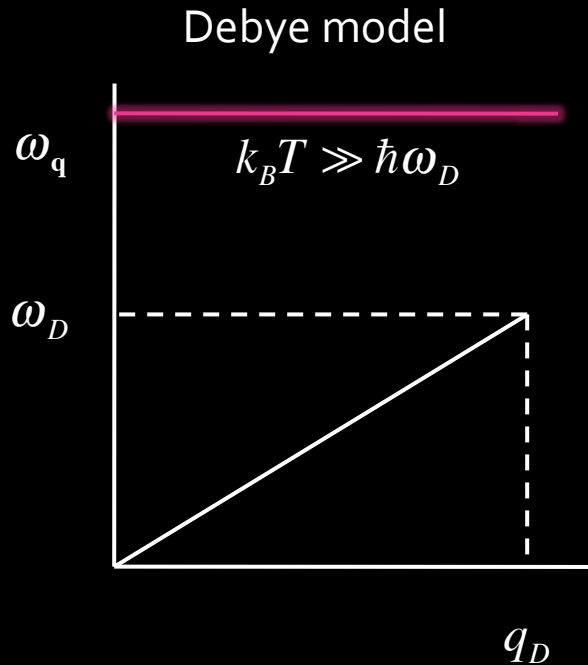
Combining the two parts together

$$\begin{aligned}
 & 2\pi \int \frac{d^3 q}{(2\pi)^3} \frac{\Xi^2 q^2}{2\rho\omega_q} \delta(\varepsilon_{\mathbf{k}+\mathbf{q}} + \hbar\omega_q - \varepsilon_{\mathbf{k}}) \\
 & \times \left[N_q f_{\mathbf{k}+\mathbf{q}} (1 - f_{\mathbf{k}}) - (N_q + 1) f_{\mathbf{k}} (1 - f_{\mathbf{k}+\mathbf{q}}) \right] \\
 & + \\
 & 2\pi \int \frac{d^3 q}{(2\pi)^3} \frac{\Xi^2 q^2}{2\rho\omega_q} \delta(\varepsilon_{\mathbf{k}+\mathbf{q}} - \hbar\omega_q - \varepsilon_{\mathbf{k}}) \\
 & \times \left[(N_q + 1) f_{\mathbf{k}+\mathbf{q}} (1 - f_{\mathbf{k}}) - N_q f_{\mathbf{k}} (1 - f_{\mathbf{k}+\mathbf{q}}) \right]
 \end{aligned}$$

After some simplifications, the Boltzmann equation reduces to

$$-e\mathbf{v} \cdot \mathbf{E} \frac{\partial f_{0\mathbf{k}}}{\partial \varepsilon_{\mathbf{k}}} = 2\pi \int \frac{d^3 q}{(2\pi)^3} \frac{\Xi^2 q^2}{2\rho\omega_q} \left[\delta(\varepsilon_{\mathbf{k}+\mathbf{q}} + \hbar\omega_q - \varepsilon_{\mathbf{k}}) \left\{ N_q (f_{\mathbf{k}+\mathbf{q}} - f_{\mathbf{k}}) - f_{\mathbf{k}} (1 - f_{\mathbf{k}+\mathbf{q}}) \right\} \right. \\
 \left. + \delta(\varepsilon_{\mathbf{k}+\mathbf{q}} - \hbar\omega_q - \varepsilon_{\mathbf{k}}) \left\{ N_q (f_{\mathbf{k}+\mathbf{q}} - f_{\mathbf{k}}) + f_{\mathbf{k}+\mathbf{q}} (1 - f_{\mathbf{k}}) \right\} \right]$$

High-temperature regime: $T \gg T_D$



$$1) N_q = \frac{1}{\exp\left(\frac{\hbar\omega_q}{k_B T}\right) - 1} \approx \frac{k_B T}{\hbar\omega_q} \geq \frac{k_B T}{\hbar\omega_D} \gg 1$$

$$2) N_q \gg 1 \text{ but } f \leq 1 \Rightarrow N_q \gg f \Rightarrow \text{neglect terms w/o } N_q$$

$$3) \text{Typical electron energies (measured from } \varepsilon_F) \\ \varepsilon_k \sim \varepsilon_{k+q} \sim k_B T \gg \hbar\omega_D \geq \hbar\omega_q \Rightarrow$$

q neglect ω_q in the δ -functions.

$$-e\mathbf{v} \cdot \mathbf{E} \frac{\partial f_{0\mathbf{k}}}{\partial \varepsilon_{\mathbf{k}}} = 2\pi \int \frac{d^3 q}{(2\pi)^3} \frac{\Xi^2 q^2}{2\rho\omega_q} \left[\delta(\varepsilon_{\mathbf{k}+\mathbf{q}} + \hbar\omega_q - \varepsilon_{\mathbf{k}}) \left\{ N_q (f_{\mathbf{k}+\mathbf{q}} - f_{\mathbf{k}}) - f_{\mathbf{k}} (1 - f_{\mathbf{k}+\mathbf{q}}) \right\} \right. \\ \left. + \delta(\varepsilon_{\mathbf{k}+\mathbf{q}} - \hbar\omega_q - \varepsilon_{\mathbf{k}}) \left\{ N_q (f_{\mathbf{k}+\mathbf{q}} - f_{\mathbf{k}}) + f_{\mathbf{k}+\mathbf{q}} (1 - f_{\mathbf{k}}) \right\} \right]$$

$$-e\mathbf{v} \cdot \mathbf{E} \frac{\partial f_{0\mathbf{k}}}{\partial \varepsilon_{\mathbf{k}}} = 2\pi \int \frac{d^3 q}{(2\pi)^3} \frac{k_B T \Xi^2 q^2}{2\hbar\rho\omega_q^2} \delta(\varepsilon_{\mathbf{k}+\mathbf{q}} - \varepsilon_{\mathbf{k}}) (f_{\mathbf{k}+\mathbf{q}} - f_{\mathbf{k}})$$

Comparison to the impurity case

Phonons:
$$-e\mathbf{v} \cdot \mathbf{E} \frac{\partial f_{0\mathbf{k}}}{\partial \varepsilon_{\mathbf{k}}} = \frac{2\pi}{\hbar} \int \frac{d^3q}{(2\pi)^3} \frac{k_B T \Xi^2 q^2}{2\rho\omega_{\mathbf{q}}^2} \delta(\varepsilon_{\mathbf{k}+\mathbf{q}} - \varepsilon_{\mathbf{k}}) (f_{\mathbf{k}+\mathbf{q}} - f_{\mathbf{k}})$$

Impurities
$$-e\mathbf{v} \cdot \mathbf{E} \frac{\partial f_{0\mathbf{k}}}{\partial \varepsilon_{\mathbf{k}}} = \frac{2\pi}{\hbar} \int \frac{d^3q}{(2\pi)^3} n_i |U(\mathbf{q})|^2 \delta(\varepsilon_{\mathbf{k}+\mathbf{q}} - \varepsilon_{\mathbf{k}}) (f_{\mathbf{k}+\mathbf{q}} - f_{\mathbf{k}})$$

$$\frac{k_B T \Xi^2 q^2}{2\rho\omega_{\mathbf{q}}^2} \Leftrightarrow n_i |U(\mathbf{q})|^2$$

$$\rho^{\text{imp}} = \frac{m}{ne^2 \tau_{\text{tr}}^{\text{imp}}} \quad \text{with} \quad \frac{1}{\tau_{\text{tr}}^{\text{imp}}} = \frac{2\pi}{\hbar} \int \frac{d^3q}{(2\pi)^3} n_i |U(\mathbf{q})|^2 \delta(\varepsilon_{\mathbf{k}+\mathbf{q}} - \varepsilon_{\mathbf{k}}) (1 - \cos \vartheta); \quad q = 2k_F \sin(\vartheta / 2)$$

$$\rho^{\text{ph}} = \frac{m}{ne^2 \tau_{\text{tr}}^{\text{ph}}} \quad \text{with} \quad \frac{1}{\tau_{\text{tr}}^{\text{ph}}} = \frac{k_B T}{\hbar} 2\pi \int_0^{q_D} \frac{d^3q}{(2\pi)^3} \frac{\Xi^2 q^2}{2\rho\omega_{\mathbf{q}}^2} \delta(\varepsilon_{\mathbf{k}+\mathbf{q}} - \varepsilon_{\mathbf{k}}) (1 - \cos \vartheta) = C \frac{k_B T}{\hbar}$$